OpenROAD Static Timing Analysis Report

Design: vsdbabysoc | Library: sky130_fd_sc_hd

Timing Corners: SS / TT / FF

SS corner: 20 ns clock, post-synthesis netlist, ideal clock, no wire parasitics (OpenSTA)

```
========== SS SETUP ==========

Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8319_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5715_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8319_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8319_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8320_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5718_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8320_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8320_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8321_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5721_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8321_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8321_/CLK (sky130_fd_sc_hd__dfxtp_1)
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  -1.08   18.92   library setup time
          18.92   data required time
-----------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
-----------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8322_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5724_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8322_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8322_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8323_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5727_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8323_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8323_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8324_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5730_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8324_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8324_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
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          18.92   data required time
         -14.73   data arrival time
-----------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8325_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5733_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8325_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8325_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8326_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5736_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8326_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8326_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8327_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5739_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8327_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8327_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
```

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           4.20   slack (MET)


Startpoint: _8191_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8328_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8191_/CLK (sky130_fd_sc_hd__dfxtp_1)
  13.47   13.47 ^ _8191_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.25   14.73 v _5742_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00   14.73 v _8328_/D (sky130_fd_sc_hd__dfxtp_1)
          14.73   data arrival time

  20.00   20.00   clock clk (rise edge)
   0.00   20.00   clock network delay (ideal)
   0.00   20.00   clock reconvergence pessimism
          20.00 ^ _8328_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   18.92   library setup time
          18.92   data required time
---------------------------------------------------------
          18.92   data required time
         -14.73   data arrival time
---------------------------------------------------------
           4.20   slack (MET)


========== SS HOLD ==========

Startpoint: _8133_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8139_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8133_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8133_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8139_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8139_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
---------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
---------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8134_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8140_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8134_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8134_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8140_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8140_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
---------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
---------------------------------------------------------
           0.64   slack (MET)
```

```
Startpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8141_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8135_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8135_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8141_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8141_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
---------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
---------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8136_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8142_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8136_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8136_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8142_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8142_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
---------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
---------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8143_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8144_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8143_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8143_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8144_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8144_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
---------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
---------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8145_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8146_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
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```
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8145_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8145_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8146_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8146_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
-----------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
-----------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8150_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8151_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8150_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8150_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8151_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8151_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
-----------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
-----------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8152_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8153_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8152_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8152_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8153_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8153_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
-----------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
-----------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8154_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8155_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8154_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8154_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8155_/D (sky130_fd_sc_hd__dfxtp_1)
```

```
          0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8155_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
-----------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
-----------------------------------------------------------
           0.64   slack (MET)


Startpoint: _8156_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8157_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

   Delay    Time   Description
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _8156_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.53    0.53 v _8156_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.53 v _8157_/D (sky130_fd_sc_hd__dfxtp_1)
           0.53   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ _8157_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.11   -0.11   library hold time
          -0.11   data required time
-----------------------------------------------------------
          -0.11   data required time
          -0.53   data arrival time
-----------------------------------------------------------
           0.64   slack (MET)


========== SS WNS/TNS ==========
worst slack 4.20
worst slack 0.64
tns 0.00
```

TT corner: 11 ns clock, post-global-route (5_1_grt.odb), propagated clock

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.

========== TT SETUP ==========
[WARNING STA-0503] report_checks -group_count is deprecated. Use -group_path_count instead.

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28    0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28    0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06    0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
           0.06    0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01    0.09    0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
           0.09    0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00    0.11    0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
           0.11    0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01    0.11    0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
           0.11    0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01    0.09    0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
           0.09    0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01    0.05    0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
           0.05    0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02    0.56    0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
           0.56    0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00    0.10    0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
           0.10    0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
```

```
   0.00    0.10    0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.10    0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03    0.18    0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.18    0.00    3.66 v _09342_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.15    0.20    3.86 ^ _09342_/Y (sky130_fd_sc_hd__a22oi_1)
           0.15    0.00    3.86 ^ _09343_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.07    0.05    3.90 v _09343_/Y (sky130_fd_sc_hd__nor2_1)
           0.07    0.00    3.90 v core.CPU_Xreg_value_a4[3][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.90   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
                   0.00   11.54   clock reconvergence pessimism
                          11.54 ^ core.CPU_Xreg_value_a4[3][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.12   11.42   library setup time
                          11.42   data required time
-----------------------------------------------------------------------
                          11.42   data required time
                          -3.90   data arrival time
-----------------------------------------------------------------------
                           7.51   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28    0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28    0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06    0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
           0.06    0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01    0.09    0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
           0.09    0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00    0.11    0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
           0.11    0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01    0.11    0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
           0.11    0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01    0.09    0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
           0.09    0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01    0.05    0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
           0.05    0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02    0.56    0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
           0.56    0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00    0.10    0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
           0.10    0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00    0.10    0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.10    0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03    0.18    0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.18    0.00    3.66 v _09433_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.15    0.20    3.86 ^ _09433_/Y (sky130_fd_sc_hd__a22oi_1)
           0.15    0.00    3.86 ^ _09434_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.07    0.05    3.90 v _09434_/Y (sky130_fd_sc_hd__nor2_1)
           0.07    0.00    3.90 v core.CPU_Xreg_value_a4[8][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.90   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
                   0.00   11.54   clock reconvergence pessimism
                          11.54 ^ core.CPU_Xreg_value_a4[8][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.12   11.42   library setup time
                          11.42   data required time
-----------------------------------------------------------------------
                          11.42   data required time
                          -3.90   data arrival time
-----------------------------------------------------------------------
                           7.51   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28    0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28    0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06    0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
           0.06    0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01    0.09    0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
```

```
          0.09    0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00    0.11    0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
           0.11    0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01    0.11    0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
           0.11    0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01    0.09    0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
           0.09    0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01    0.05    0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
           0.05    0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02    0.56    0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
           0.56    0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00    0.10    0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
           0.10    0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00    0.10    0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.10    0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03    0.18    0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.18    0.00    3.66 v _09251_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.15    0.20    3.86 ^ _09251_/Y (sky130_fd_sc_hd__a22oi_1)
           0.15    0.00    3.86 ^ _09252_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.07    0.05    3.90 v _09252_/Y (sky130_fd_sc_hd__nor2_1)
           0.07    0.00    3.90 v core.CPU_Xreg_value_a4[2][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.90   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
                   0.00   11.54   clock reconvergence pessimism
                          11.54 ^ core.CPU_Xreg_value_a4[2][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.12   11.42   library setup time
                          11.42   data required time
-----------------------------------------------------------------------------
                          11.42   data required time
                          -3.90   data arrival time
-----------------------------------------------------------------------------
                           7.51   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28    0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28    0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06    0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
           0.06    0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01    0.09    0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
           0.09    0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00    0.11    0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
           0.11    0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01    0.11    0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
           0.11    0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01    0.09    0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
           0.09    0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01    0.05    0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
           0.05    0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02    0.56    0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
           0.56    0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00    0.10    0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
           0.10    0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00    0.10    0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.10    0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03    0.18    0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.18    0.00    3.66 v _09164_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.15    0.20    3.86 ^ _09164_/Y (sky130_fd_sc_hd__a22oi_1)
           0.15    0.00    3.86 ^ _09165_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.07    0.05    3.90 v _09165_/Y (sky130_fd_sc_hd__nor2_1)
           0.07    0.00    3.90 v core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.90   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
                   0.00   11.54   clock reconvergence pessimism
                          11.54 ^ core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.12   11.42   library setup time
                          11.42   data required time
-----------------------------------------------------------------------------
                          11.42   data required time
                          -3.90   data arrival time
-----------------------------------------------------------------------------
                           7.51   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[9][21]$_SDFFE_PP0P_
```

```
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
            0.05   0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03     0.28   0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
            0.28   0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00     0.06   0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
            0.06   0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01     0.09   0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
            0.09   0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00     0.11   0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
            0.11   0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01     0.11   0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
            0.11   0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01     0.09   0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
            0.09   0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01     0.05   0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
            0.05   0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02     0.56   0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
            0.56   0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00     0.10   0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
            0.10   0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00     0.10   0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
            0.10   0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03     0.18   0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
            0.18   0.00    3.66 v _09519_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00     0.15   0.20    3.86 ^ _09519_/Y (sky130_fd_sc_hd__a22oi_1)
            0.15   0.00    3.86 ^ _09520_/B (sky130_fd_sc_hd__nor2_1)
   0.00     0.07   0.05    3.90 v _09520_/Y (sky130_fd_sc_hd__nor2_1)
            0.07   0.00    3.90 v core.CPU_Xreg_value_a4[9][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.90   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
                   0.00   11.54   clock reconvergence pessimism
                          11.54 ^ core.CPU_Xreg_value_a4[9][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.12   11.42   library setup time
                          11.42   data required time
---------------------------------------------------------------------------
                          11.42   data required time
                          -3.90   data arrival time
---------------------------------------------------------------------------
                           7.51   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
           (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
            0.05   0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03     0.28   0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
            0.28   0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00     0.06   0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
            0.06   0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01     0.09   0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
            0.09   0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00     0.11   0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
            0.11   0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01     0.11   0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
            0.11   0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01     0.09   0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
            0.09   0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01     0.05   0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
            0.05   0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02     0.56   0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
            0.56   0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00     0.10   0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
            0.10   0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00     0.10   0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
            0.10   0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03     0.18   0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
            0.18   0.00    3.66 v _08785_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00     0.12   0.18    3.84 ^ _08785_/Y (sky130_fd_sc_hd__a22oi_2)
            0.12   0.00    3.84 ^ _08786_/B (sky130_fd_sc_hd__nor2_1)
   0.00     0.07   0.05    3.89 v _08786_/Y (sky130_fd_sc_hd__nor2_1)
            0.07   0.00    3.89 v core.CPU_Xreg_value_a4[16][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.89   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
```

```
                  0.00   11.54   clock reconvergence pessimism
                         11.54 ^ core.CPU_Xreg_value_a4[16][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.12   11.42   library setup time
                         11.42   data required time
-----------------------------------------------------------------------------
                         11.42   data required time
                         -3.89   data arrival time
-----------------------------------------------------------------------------
                          7.53   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[26][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.53    0.53   clock network delay (propagated)
           0.05   0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28   0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28   0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06   0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
           0.06   0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01    0.09   0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
           0.09   0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00    0.11   0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
           0.11   0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01    0.11   0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
           0.11   0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01    0.09   0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
           0.09   0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01    0.05   0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
           0.05   0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02    0.56   0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
           0.56   0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00    0.10   0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
           0.10   0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00    0.10   0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.10   0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03    0.18   0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.18   0.00    3.66 v _09076_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.12   0.18    3.84 ^ _09076_/Y (sky130_fd_sc_hd__a22oi_2)
           0.12   0.00    3.84 ^ _09077_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.07   0.05    3.89 v _09077_/Y (sky130_fd_sc_hd__nor2_1)
           0.07   0.00    3.89 v core.CPU_Xreg_value_a4[26][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                          3.89   data arrival time
                 11.00   11.00   clock clk (rise edge)
                  0.54   11.54   clock network delay (propagated)
                  0.00   11.54   clock reconvergence pessimism
                         11.54 ^ core.CPU_Xreg_value_a4[26][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.12   11.42   library setup time
                         11.42   data required time
-----------------------------------------------------------------------------
                         11.42   data required time
                         -3.89   data arrival time
-----------------------------------------------------------------------------
                          7.53   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.53    0.53   clock network delay (propagated)
           0.05   0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28   0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28   0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06   0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
           0.06   0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01    0.09   0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
           0.09   0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00    0.11   0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
           0.11   0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01    0.11   0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
           0.11   0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01    0.09   0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
           0.09   0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01    0.05   0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
           0.05   0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02    0.56   0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
           0.56   0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
```

```
   0.00    0.10    0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
           0.10    0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00    0.10    0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.10    0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03    0.18    0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.18    0.00    3.66 v _08879_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.12    0.18    3.84 ^ _08879_/Y (sky130_fd_sc_hd__a22oi_2)
           0.12    0.00    3.84 ^ _08880_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.07    0.05    3.89 v _08880_/Y (sky130_fd_sc_hd__nor2_1)
           0.07    0.00    3.89 v core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.89   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
                   0.00   11.54   clock reconvergence pessimism
                          11.54 ^ core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.12   11.42   library setup time
                          11.42   data required time
-----------------------------------------------------------------------
                          11.42   data required time
                          -3.89   data arrival time
-----------------------------------------------------------------------
                           7.53   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28    0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28    0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06    0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
           0.06    0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01    0.09    0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
           0.09    0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00    0.11    0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
           0.11    0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01    0.11    0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
           0.11    0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01    0.09    0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
           0.09    0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01    0.05    0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
           0.05    0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02    0.56    0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
           0.56    0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00    0.10    0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
           0.10    0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00    0.10    0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.10    0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03    0.18    0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.18    0.00    3.66 v _08674_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.12    0.18    3.84 ^ _08674_/Y (sky130_fd_sc_hd__a22oi_2)
           0.12    0.00    3.84 ^ _08675_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.07    0.05    3.89 v _08675_/Y (sky130_fd_sc_hd__nor2_1)
           0.07    0.00    3.89 v core.CPU_Xreg_value_a4[11][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.89   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.54   11.54   clock network delay (propagated)
                   0.00   11.54   clock reconvergence pessimism
                          11.54 ^ core.CPU_Xreg_value_a4[11][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.12   11.42   library setup time
                          11.42   data required time
-----------------------------------------------------------------------
                          11.42   data required time
                          -3.89   data arrival time
-----------------------------------------------------------------------
                           7.53   slack (MET)

Startpoint: core.CPU_src2_value_a3[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_src2_value_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.03    0.28    0.48    1.00 ^ core.CPU_src2_value_a3[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.28    0.00    1.00 ^ _05409_/A (sky130_fd_sc_hd__inv_1)
   0.00    0.06    0.07    1.08 v _05409_/Y (sky130_fd_sc_hd__inv_1)
```

```
          0.06    0.00    1.08 v _10740_/B (sky130_fd_sc_hd__ha_1)
   0.01   0.09    0.31    1.39 ^ _10740_/SUM (sky130_fd_sc_hd__ha_1)
          0.09    0.00    1.39 ^ _07488_/B_N (sky130_fd_sc_hd__nor2b_1)
   0.00   0.11    0.15    1.54 ^ _07488_/Y (sky130_fd_sc_hd__nor2b_1)
          0.11    0.00    1.54 ^ _07489_/A2 (sky130_fd_sc_hd__o2111a_1)
   0.01   0.11    0.23    1.77 ^ _07489_/X (sky130_fd_sc_hd__o2111a_1)
          0.11    0.00    1.77 ^ _07617_/A2 (sky130_fd_sc_hd__o21ai_1)
   0.01   0.09    0.10    1.87 v _07617_/Y (sky130_fd_sc_hd__o21ai_1)
          0.09    0.00    1.87 v _07711_/A3 (sky130_fd_sc_hd__a31o_2)
   0.01   0.05    0.27    2.14 v _07711_/X (sky130_fd_sc_hd__a31o_2)
          0.05    0.00    2.14 v _07964_/A2 (sky130_fd_sc_hd__a211oi_1)
   0.02   0.56    0.50    2.64 ^ _07964_/Y (sky130_fd_sc_hd__a211oi_1)
          0.56    0.00    2.64 ^ _07978_/B (sky130_fd_sc_hd__nor4b_1)
   0.00   0.10    0.10    2.75 v _07978_/Y (sky130_fd_sc_hd__nor4b_1)
          0.10    0.00    2.75 v _07987_/C (sky130_fd_sc_hd__or4_1)
   0.00   0.10    0.49    3.24 v _07987_/X (sky130_fd_sc_hd__or4_1)
          0.10    0.00    3.24 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.03   0.18    0.42    3.66 v _07988_/X (sky130_fd_sc_hd__a211o_1)
          0.18    0.00    3.66 v _08977_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00   0.12    0.18    3.84 ^ _08977_/Y (sky130_fd_sc_hd__a22oi_2)
          0.12    0.00    3.84 ^ _08978_/B (sky130_fd_sc_hd__nor2_1)
   0.00   0.07    0.05    3.89 v _08978_/Y (sky130_fd_sc_hd__nor2_1)
          0.07    0.00    3.89 v core.CPU_Xreg_value_a4[1][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                          3.89   data arrival time
                 11.00   11.00   clock clk (rise edge)
                  0.54   11.54   clock network delay (propagated)
                  0.00   11.54   clock reconvergence pessimism
                         11.54 ^ core.CPU_Xreg_value_a4[1][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.12   11.42   library setup time
                         11.42   data required time
-----------------------------------------------------------------------------
                         11.42   data required time
                         -3.89   data arrival time
-----------------------------------------------------------------------------
                          7.53   slack (MET)

========== TT HOLD ==========
[WARNING STA-0503] report_checks -group_count is deprecated. Use -group_path_count instead.

Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
          0.05     0.00    0.53 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.03     0.29    0.83 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03     0.00    0.83 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
                           0.83   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
                   0.00    0.53   clock reconvergence pessimism
                           0.53 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
                  -0.03    0.51   library hold time
                           0.51   data required time
-----------------------------------------------------------------------------
                           0.51   data required time
                          -0.83   data arrival time
-----------------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_rd_valid_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_valid_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.52    0.52   clock network delay (propagated)
          0.05     0.00    0.52 ^ core.CPU_rd_valid_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.03     0.29    0.82 ^ core.CPU_rd_valid_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03     0.00    0.82 ^ core.CPU_rd_valid_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.52    0.52   clock network delay (propagated)
                   0.00    0.52   clock reconvergence pessimism
                           0.52 ^ core.CPU_rd_valid_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.49   library hold time
                           0.49   data required time
-----------------------------------------------------------------------------
                           0.49   data required time
```

```
                          -0.82   data arrival time
-----------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.29    0.82 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.82 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
                   0.00    0.53   clock reconvergence pessimism
                           0.53 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.50   library hold time
                           0.50   data required time
---------------------------------------------------------------------
                           0.50   data required time
                          -0.82   data arrival time
---------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_is_sll_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_sll_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_is_sll_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.29    0.82 ^ core.CPU_is_sll_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.82 ^ core.CPU_is_sll_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
                   0.00    0.53   clock reconvergence pessimism
                           0.53 ^ core.CPU_is_sll_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.50   library hold time
                           0.50   data required time
---------------------------------------------------------------------
                           0.50   data required time
                          -0.82   data arrival time
---------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_is_slli_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_slli_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_is_slli_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.29    0.82 ^ core.CPU_is_slli_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.82 ^ core.CPU_is_slli_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
                   0.00    0.53   clock reconvergence pessimism
                           0.53 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.50   library hold time
                           0.50   data required time
---------------------------------------------------------------------
                           0.50   data required time
                          -0.82   data arrival time
---------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_inc_pc_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
```

```
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.29    0.82 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.82 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
                   0.00    0.53   clock reconvergence pessimism
                           0.53 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.50   library hold time
                           0.50   data required time
---------------------------------------------------------------------------
                           0.50   data required time
                          -0.82   data arrival time
---------------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_imm_a2[10]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[10]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_imm_a2[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.29    0.82 ^ core.CPU_imm_a2[10]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.82 ^ core.CPU_imm_a3[10]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
                   0.00    0.53   clock reconvergence pessimism
                           0.53 ^ core.CPU_imm_a3[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.50   library hold time
                           0.50   data required time
---------------------------------------------------------------------------
                           0.50   data required time
                          -0.82   data arrival time
---------------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_inc_pc_a2[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[2]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.29    0.82 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.82 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
                   0.00    0.53   clock reconvergence pessimism
                           0.53 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.50   library hold time
                           0.50   data required time
---------------------------------------------------------------------------
                           0.50   data required time
                          -0.82   data arrival time
---------------------------------------------------------------------------
                           0.32   slack (MET)

Startpoint: core.CPU_inc_pc_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.53    0.53   clock network delay (propagated)
           0.05    0.00    0.53 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.29    0.82 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.82 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.82   data arrival time
```

```
                  0.00    0.00   clock clk (rise edge)
                  0.53    0.53   clock network delay (propagated)
                  0.00    0.53   clock reconvergence pessimism
                          0.53 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.03    0.50   library hold time
                          0.50   data required time
-----------------------------------------------------------------------------
                          0.50   data required time
                         -0.82   data arrival time
-----------------------------------------------------------------------------
                          0.32   slack (MET)

Startpoint: core.CPU_inc_pc_a2[5]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[5]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.53    0.53   clock network delay (propagated)
           0.05   0.00    0.53 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03   0.29    0.82 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03   0.00    0.82 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.82   data arrival time
                  0.00    0.00   clock clk (rise edge)
                  0.53    0.53   clock network delay (propagated)
                  0.00    0.53   clock reconvergence pessimism
                          0.53 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.03    0.50   library hold time
                          0.50   data required time
-----------------------------------------------------------------------------
                          0.50   data required time
                         -0.82   data arrival time
-----------------------------------------------------------------------------
                          0.32   slack (MET)

========== TT WNS/TNS ==========
worst slack max 7.51
worst slack min 0.32
tns max 0.00
tns min 0.00
```

FF corner: 11 ns clock, post-global-route (5_1_grt.odb), propagated clock

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.

========== FF SETUP ==========
[WARNING STA-0503] report_checks -group_count is deprecated. Use -group_path_count instead.

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
           0.04   0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03   0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03   0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15   0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15   0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11   0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11   0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36   0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36   0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07   0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07   0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13   0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13   0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04   0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04   0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04   0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04   0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03   0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03   0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07   0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07   0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16   0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
```

```
          0.16    0.00    2.26 v _09342_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.09    0.10    2.37 ^ _09342_/Y (sky130_fd_sc_hd__a22oi_1)
          0.09    0.00    2.37 ^ _09343_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.39 v _09343_/Y (sky130_fd_sc_hd__nor2_1)
          0.06    0.00    2.39 v core.CPU_Xreg_value_a4[3][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                          2.39   data arrival time
                 11.00   11.00   clock clk (rise edge)
                  0.38   11.38   clock network delay (propagated)
                  0.00   11.38   clock reconvergence pessimism
                         11.38 ^ core.CPU_Xreg_value_a4[3][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.07   11.32   library setup time
                         11.32   data required time
-----------------------------------------------------------------------
                         11.32   data required time
                         -2.39   data arrival time
-----------------------------------------------------------------------
                          8.93   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
          0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
          0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
          0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
          0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
          0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
          0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
          0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
          0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
          0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
          0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
          0.16    0.00    2.26 v _09433_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.09    0.10    2.37 ^ _09433_/Y (sky130_fd_sc_hd__a22oi_1)
          0.09    0.00    2.37 ^ _09434_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.39 v _09434_/Y (sky130_fd_sc_hd__nor2_1)
          0.06    0.00    2.39 v core.CPU_Xreg_value_a4[8][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                          2.39   data arrival time
                 11.00   11.00   clock clk (rise edge)
                  0.38   11.38   clock network delay (propagated)
                  0.00   11.38   clock reconvergence pessimism
                         11.38 ^ core.CPU_Xreg_value_a4[8][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.07   11.32   library setup time
                         11.32   data required time
-----------------------------------------------------------------------
                         11.32   data required time
                         -2.39   data arrival time
-----------------------------------------------------------------------
                          8.93   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
          0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
          0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
          0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
          0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
```

```
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _09251_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.09    0.10    2.37 ^ _09251_/Y (sky130_fd_sc_hd__a22oi_1)
           0.09    0.00    2.37 ^ _09252_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.39 v _09252_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.39 v core.CPU_Xreg_value_a4[2][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.39   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[2][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
                          11.32   data required time
-----------------------------------------------------------------------------
                          11.32   data required time
                          -2.39   data arrival time
-----------------------------------------------------------------------------
                           8.93   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _09164_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.09    0.10    2.37 ^ _09164_/Y (sky130_fd_sc_hd__a22oi_1)
           0.09    0.00    2.37 ^ _09165_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.39 v _09165_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.39 v core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.39   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[27][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
                          11.32   data required time
-----------------------------------------------------------------------------
                          11.32   data required time
                          -2.39   data arrival time
-----------------------------------------------------------------------------
                           8.93   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[9][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
```

```
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _09519_/B2 (sky130_fd_sc_hd__a22oi_1)
   0.00    0.09    0.10    2.37 ^ _09519_/Y (sky130_fd_sc_hd__a22oi_1)
           0.09    0.00    2.37 ^ _09520_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.39 v _09520_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.39 v core.CPU_Xreg_value_a4[9][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.39   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[9][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
                          11.32   data required time
---------------------------------------------------------------------------
                          11.32   data required time
                          -2.39   data arrival time
---------------------------------------------------------------------------
                           8.93   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _08785_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.07    0.09    2.35 ^ _08785_/Y (sky130_fd_sc_hd__a22oi_2)
           0.07    0.00    2.35 ^ _08786_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.38 v _08786_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.38 v core.CPU_Xreg_value_a4[16][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.38   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[16][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
```

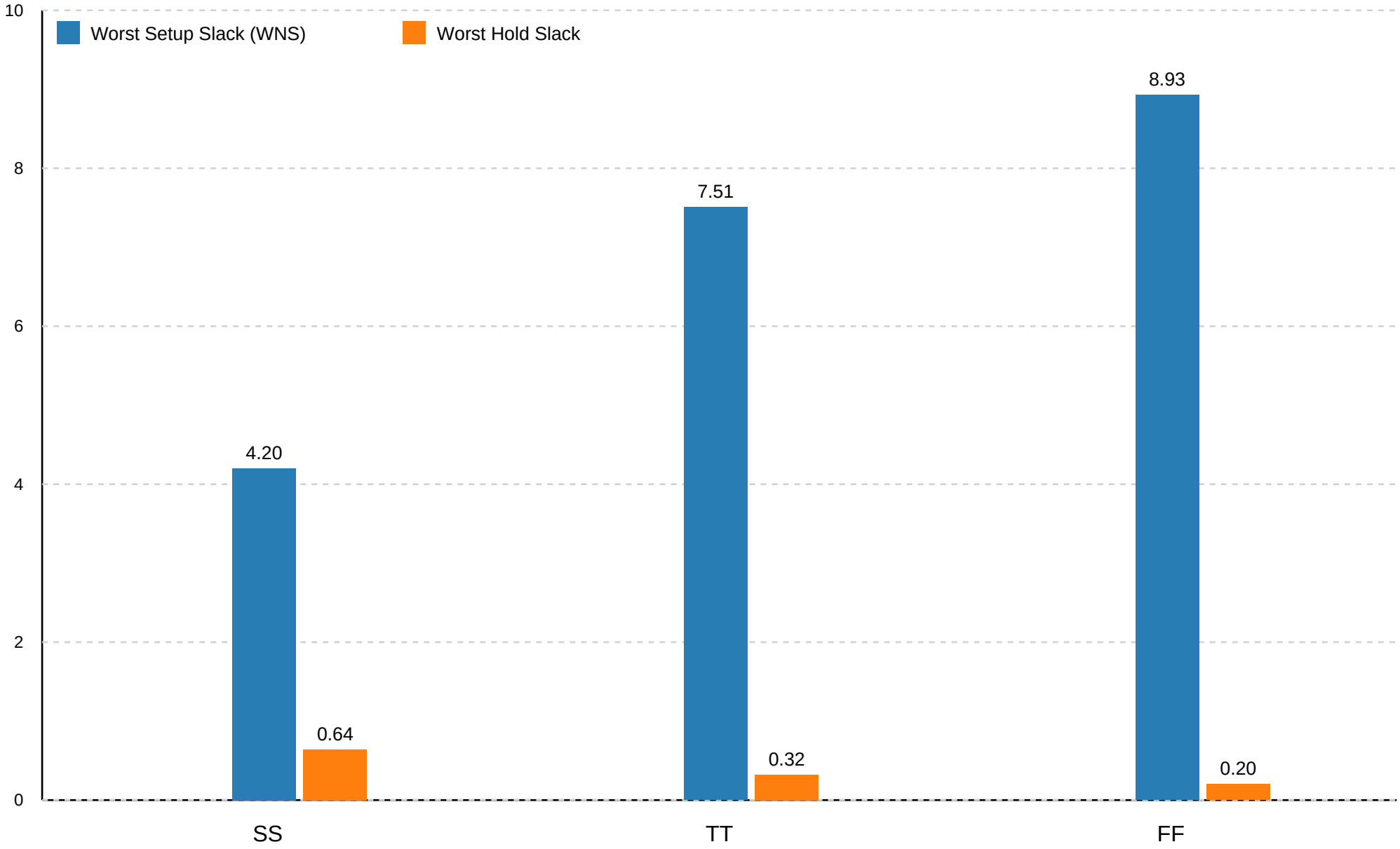
```
                          11.32   data required time
-----------------------------------------------------------------------------
                          11.32   data required time
                          -2.38   data arrival time
-----------------------------------------------------------------------------
                           8.94   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[26][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _09076_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.07    0.09    2.35 ^ _09076_/Y (sky130_fd_sc_hd__a22oi_2)
           0.07    0.00    2.35 ^ _09077_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.38 v _09077_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.38 v core.CPU_Xreg_value_a4[26][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.38   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[26][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
                          11.32   data required time
-----------------------------------------------------------------------------
                          11.32   data required time
                          -2.38   data arrival time
-----------------------------------------------------------------------------
                           8.94   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
```

```
          0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _08879_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.07    0.09    2.35 ^ _08879_/Y (sky130_fd_sc_hd__a22oi_2)
           0.07    0.00    2.35 ^ _08880_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.38 v _08880_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.38 v core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.38   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
                          11.32   data required time
-----------------------------------------------------------------------------
                          11.32   data required time
                          -2.38   data arrival time
-----------------------------------------------------------------------------
                           8.94   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _08674_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.07    0.09    2.35 ^ _08674_/Y (sky130_fd_sc_hd__a22oi_2)
           0.07    0.00    2.35 ^ _08675_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.38 v _08675_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.38 v core.CPU_Xreg_value_a4[11][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.38   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[11][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
                          11.32   data required time
-----------------------------------------------------------------------------
                          11.32   data required time
                          -2.38   data arrival time
-----------------------------------------------------------------------------
                           8.94   slack (MET)

Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][21]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ place471/A (sky130_fd_sc_hd__buf_4)
   0.08    0.15    0.16    0.73 ^ place471/X (sky130_fd_sc_hd__buf_4)
           0.15    0.00    0.73 ^ _07381_/S (sky130_fd_sc_hd__mux2i_4)
   0.00    0.11    0.15    0.88 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
           0.11    0.00    0.88 ^ place415/A (sky130_fd_sc_hd__buf_4)
```

```
   0.19    0.36    0.32    1.19 ^ place415/X (sky130_fd_sc_hd__buf_4)
           0.36    0.00    1.19 ^ _07591_/S0 (sky130_fd_sc_hd__mux4_2)
   0.00    0.07    0.33    1.52 v _07591_/X (sky130_fd_sc_hd__mux4_2)
           0.07    0.00    1.52 v _07592_/A1 (sky130_fd_sc_hd__mux2i_1)
   0.01    0.13    0.09    1.61 ^ _07592_/Y (sky130_fd_sc_hd__mux2i_1)
           0.13    0.00    1.61 ^ _07972_/B (sky130_fd_sc_hd__nand2_1)
   0.00    0.04    0.04    1.65 v _07972_/Y (sky130_fd_sc_hd__nand2_1)
           0.04    0.00    1.65 v _07973_/C (sky130_fd_sc_hd__nand3_1)
   0.00    0.04    0.05    1.70 ^ _07973_/Y (sky130_fd_sc_hd__nand3_1)
           0.04    0.00    1.70 ^ _07975_/A1 (sky130_fd_sc_hd__a21oi_1)
   0.00    0.03    0.04    1.74 v _07975_/Y (sky130_fd_sc_hd__a21oi_1)
           0.03    0.00    1.74 v _07987_/A (sky130_fd_sc_hd__or4_1)
   0.00    0.07    0.29    2.03 v _07987_/X (sky130_fd_sc_hd__or4_1)
           0.07    0.00    2.03 v _07988_/B1 (sky130_fd_sc_hd__a211o_1)
   0.04    0.16    0.23    2.26 v _07988_/X (sky130_fd_sc_hd__a211o_1)
           0.16    0.00    2.26 v _08977_/B2 (sky130_fd_sc_hd__a22oi_2)
   0.00    0.07    0.09    2.35 ^ _08977_/Y (sky130_fd_sc_hd__a22oi_2)
           0.07    0.00    2.35 ^ _08978_/B (sky130_fd_sc_hd__nor2_1)
   0.00    0.06    0.02    2.38 v _08978_/Y (sky130_fd_sc_hd__nor2_1)
           0.06    0.00    2.38 v core.CPU_Xreg_value_a4[1][21]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           2.38   data arrival time
                  11.00   11.00   clock clk (rise edge)
                   0.38   11.38   clock network delay (propagated)
                   0.00   11.38   clock reconvergence pessimism
                          11.38 ^ core.CPU_Xreg_value_a4[1][21]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.07   11.32   library setup time
                          11.32   data required time
-----------------------------------------------------------------------------
                          11.32   data required time
                          -2.38   data arrival time
-----------------------------------------------------------------------------
                           8.94   slack (MET)

========== FF HOLD ==========
[WARNING STA-0503] report_checks -group_count is deprecated. Use -group_path_count instead.

Startpoint: core.CPU_rd_valid_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_valid_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.37    0.37   clock network delay (propagated)
           0.04    0.00    0.37 ^ core.CPU_rd_valid_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.56 ^ core.CPU_rd_valid_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.56 ^ core.CPU_rd_valid_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.56   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.37    0.37   clock network delay (propagated)
                   0.00    0.37   clock reconvergence pessimism
                           0.37 ^ core.CPU_rd_valid_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.36   library hold time
                           0.36   data required time
-----------------------------------------------------------------------------
                           0.36   data required time
                          -0.56   data arrival time
-----------------------------------------------------------------------------
                           0.20   slack (MET)

Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.56 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.56 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.56   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
                   0.00    0.38   clock reconvergence pessimism
                           0.38 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.36   library hold time
                           0.36   data required time
-----------------------------------------------------------------------------
                           0.36   data required time
                          -0.56   data arrival time
-----------------------------------------------------------------------------
                           0.20   slack (MET)
```

```
Startpoint: core.CPU_is_sll_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_sll_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_sll_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.56 ^ core.CPU_is_sll_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.56 ^ core.CPU_is_sll_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.56   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
                   0.00    0.38   clock reconvergence pessimism
                           0.38 ^ core.CPU_is_sll_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.36   library hold time
                           0.36   data required time
---------------------------------------------------------------------------
                           0.36   data required time
                          -0.56   data arrival time
---------------------------------------------------------------------------
                           0.20   slack (MET)

Startpoint: core.CPU_is_slli_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_slli_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_is_slli_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.56 ^ core.CPU_is_slli_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.56 ^ core.CPU_is_slli_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.56   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
                   0.00    0.38   clock reconvergence pessimism
                           0.38 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.36   library hold time
                           0.36   data required time
---------------------------------------------------------------------------
                           0.36   data required time
                          -0.56   data arrival time
---------------------------------------------------------------------------
                           0.20   slack (MET)

Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
                           0.57   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
                   0.00    0.38   clock reconvergence pessimism
                           0.38 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
                  -0.02    0.37   library hold time
                           0.37   data required time
---------------------------------------------------------------------------
                           0.37   data required time
                          -0.57   data arrival time
---------------------------------------------------------------------------
                           0.20   slack (MET)

Startpoint: core.CPU_inc_pc_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
```

```
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
          0.04    0.00    0.38 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.03    0.19    0.57 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03    0.00    0.57 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.57   data arrival time
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
                  0.00    0.38   clock reconvergence pessimism
                          0.38 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.36   library hold time
                          0.36   data required time
-----------------------------------------------------------------------
                          0.36   data required time
                         -0.57   data arrival time
-----------------------------------------------------------------------
                          0.20   slack (MET)

Startpoint: core.CPU_imm_a2[10]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[10]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
          0.04    0.00    0.38 ^ core.CPU_imm_a2[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.03    0.19    0.57 ^ core.CPU_imm_a2[10]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03    0.00    0.57 ^ core.CPU_imm_a3[10]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.57   data arrival time
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
                  0.00    0.38   clock reconvergence pessimism
                          0.38 ^ core.CPU_imm_a3[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.36   library hold time
                          0.36   data required time
-----------------------------------------------------------------------
                          0.36   data required time
                         -0.57   data arrival time
-----------------------------------------------------------------------
                          0.20   slack (MET)

Startpoint: core.CPU_inc_pc_a2[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[2]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
          0.04    0.00    0.38 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.03    0.19    0.57 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03    0.00    0.57 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.57   data arrival time
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
                  0.00    0.38   clock reconvergence pessimism
                          0.38 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.36   library hold time
                          0.36   data required time
-----------------------------------------------------------------------
                          0.36   data required time
                         -0.57   data arrival time
-----------------------------------------------------------------------
                          0.20   slack (MET)

Startpoint: core.CPU_inc_pc_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
          0.04    0.00    0.38 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.03    0.19    0.57 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03    0.00    0.57 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.57   data arrival time
                  0.00    0.00   clock clk (rise edge)
                  0.38    0.38   clock network delay (propagated)
                  0.00    0.38   clock reconvergence pessimism
```

```
                          0.38 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.36   library hold time
                          0.36   data required time
-----------------------------------------------------------------------------
                          0.36   data required time
                         -0.57   data arrival time
-----------------------------------------------------------------------------
                          0.20   slack (MET)

Startpoint: core.CPU_inc_pc_a2[5]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[5]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
           0.04    0.00    0.38 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.57 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.57 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.57   data arrival time
                   0.00    0.00   clock clk (rise edge)
                   0.38    0.38   clock network delay (propagated)
                   0.00    0.38   clock reconvergence pessimism
                           0.38 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.36   library hold time
                           0.36   data required time
-----------------------------------------------------------------------------
                           0.36   data required time
                          -0.57   data arrival time
-----------------------------------------------------------------------------
                           0.20   slack (MET)

========== FF WNS/TNS ==========
worst slack max 8.93
worst slack min 0.20
tns max 0.00
tns min 0.00
```

STA Timing Slack Across SS / TT / FF Corners